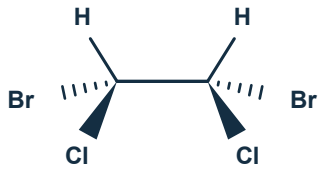
CONFORMATIONS LAB

Rotate. Look. Project.

CHClBr-CHClBr / Newman projections



Start: Cl forward; Br behind

Fixed starting arrangement; only C-C bond rotation changes between views.

1 Look down C1 -> C2. 2 Fill each box with H, Cl, or Br.

3 Classify each view as eclipsed or staggered. Use each group once per carbon.

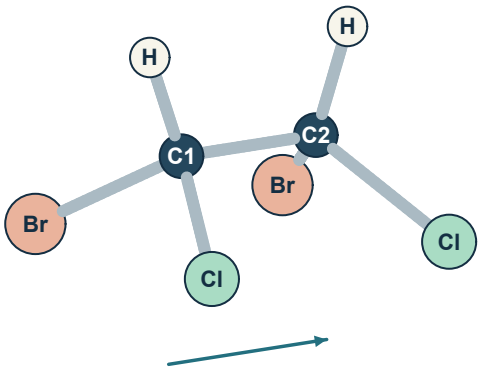
Point = front C1 Circle = back C2 A-C = front groups D-F = back groups

Eclipsed rear bonds are offset slightly for clarity. Molecular angles are unchanged.

01 H-C1-C2-H rotation: 0 degrees

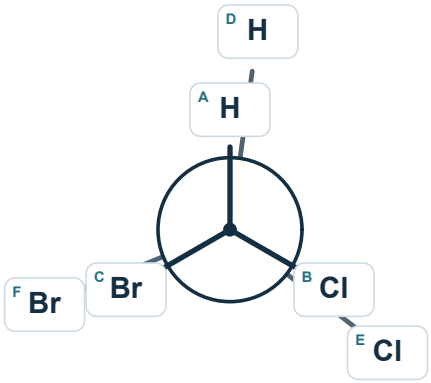
ECLIPSED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW

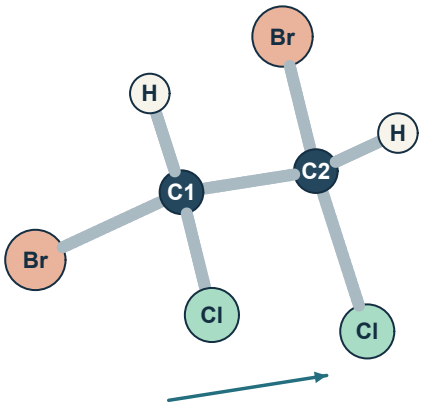


C1 in front | C2 at back

02 H-C1-C2-H rotation: 60 degrees

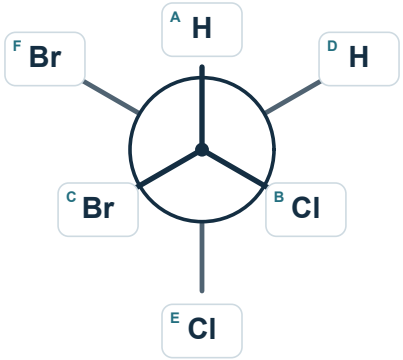
STAGGERED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW

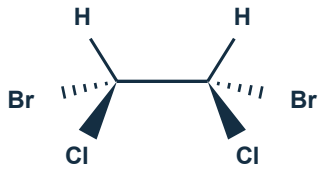


C1 in front | C2 at back

CONFORMATIONS LAB

Rotate. Look. Project.

CHClBr-CHClBr / Newman projections



Start: Cl forward; Br behind

Fixed starting arrangement; only C-C bond rotation changes between views.

1 Look down C1 -> C2. 2 Fill each box with H, Cl, or Br.

3 Classify each view as eclipsed or staggered. Use each group once per carbon.

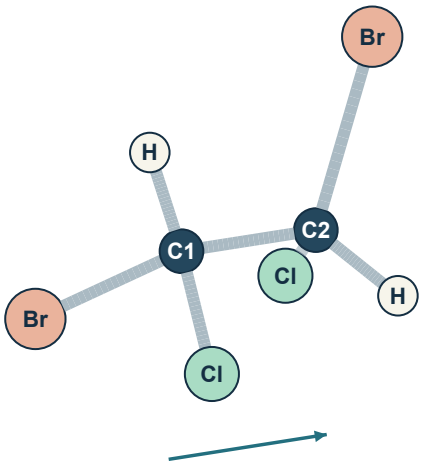
Point = front C1 Circle = back C2 A-C = front groups D-F = back groups

Eclipsed rear bonds are offset slightly for clarity. Molecular angles are unchanged.

03 H-C1-C2-H rotation: 120 degrees

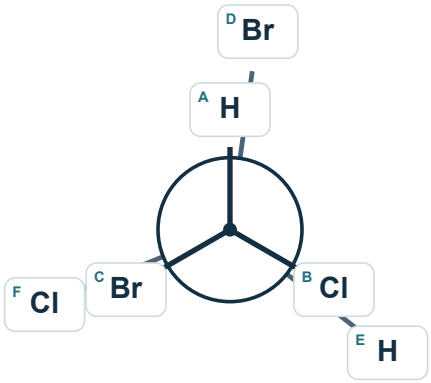
ECLIPSED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW

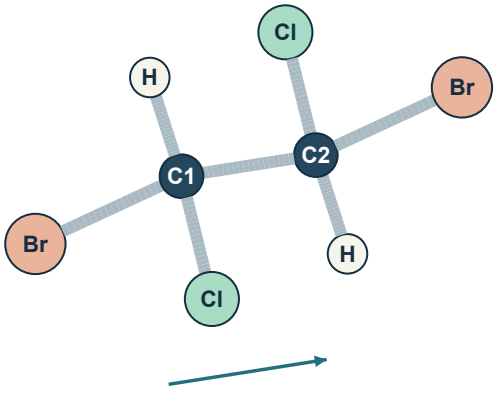


C1 in front | C2 at back

04 H-C1-C2-H rotation: 180 degrees

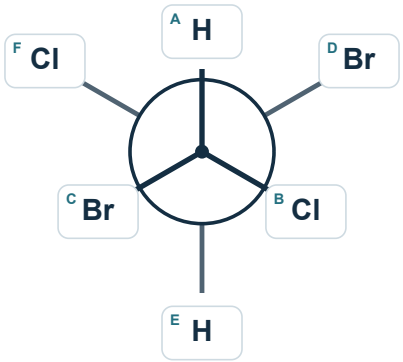
STAGGERED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW

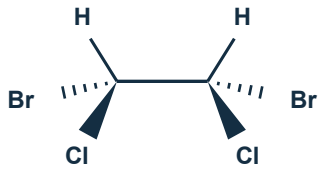


C1 in front | C2 at back

CONFORMATIONS LAB

Rotate. Look. Project.

CHClBr-CHClBr / Newman projections



Start: Cl forward; Br behind

Fixed starting arrangement; only C-C bond rotation changes between views.

1 Look down C1 -> C2. 2 Fill each box with H, Cl, or Br.

3 Classify each view as eclipsed or staggered. Use each group once per carbon.

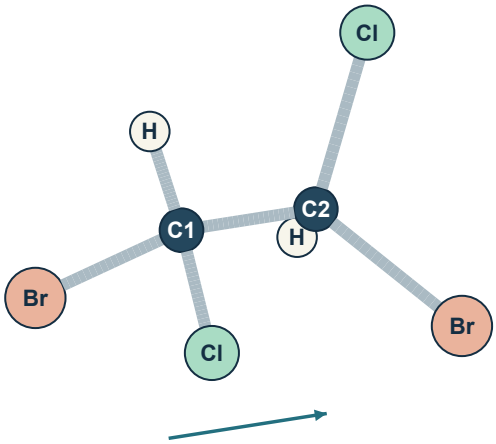
Point = front C1 Circle = back C2 A-C = front groups D-F = back groups

Eclipsed rear bonds are offset slightly for clarity. Molecular angles are unchanged.

05 H-C1-C2-H rotation: 240 degrees

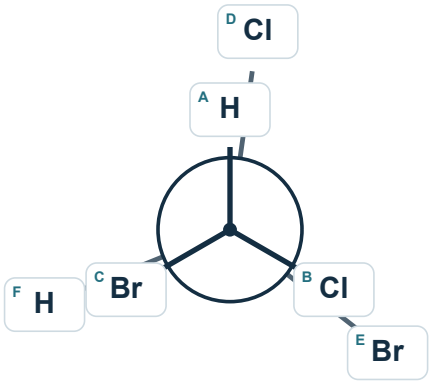
ECLIPSED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW

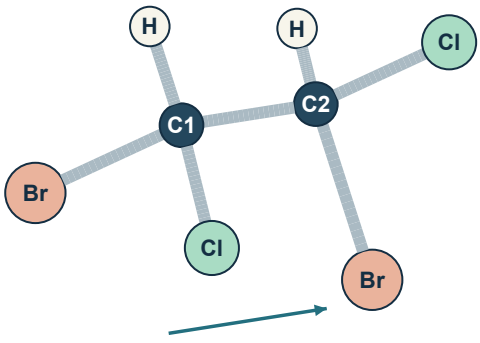


C1 in front | C2 at back

06 H-C1-C2-H rotation: 300 degrees

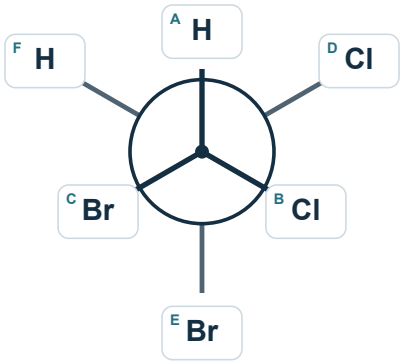
STAGGERED

OBLIQUE MOLECULAR VIEW



Look from C1 toward C2

MATCHING NEWMAN VIEW



C1 in front | C2 at back